Memory Is the Defence: Physics-Gated Continuum-Memory Receivers for Poisoning-Resilient Test-Time Learning

Abraheem Rashid

***Abstract*—Neural receivers that learn at test time adapt to the channel from the pilots of every coherence block, so the pilot stream is a training set that an adversary can write to. This paper shows that such a receiver resists pilot poisoning when it consolidates into persistent memory only what a label-free physical check verifies, and when that check judges every update against the protected memory rather than against a reference the attacker also reaches. Citadel-Rx realises the principle in a DeepSIC receiver with a fast level fitted on every block and two slow levels updated every 8 and 32 blocks through a gated, clipped average of the faster states; whatever the attacker does, the clip bounds how far the slowest level can be moved. A receiver with the same CRC test and rollback but a single memory fails: its reference, LMMSE on the poisoned pilots, degrades with the attack, so the first poisoned block is accepted on every seed and becomes the state it rolls back to. With all settings fixed on calibration seeds and results on ten held-out seeds, white-box poisoning at 0 dB signal-to-interference ratio raises the symbol error rate of Citadel-Rx by $+0.0188$ over its own clean run, against $+0.768$ for tuned online DeepSIC, and 99% of the poisoned blocks are rejected at a 0.1% false-rejection rate. Attacks built against the gates do not raise this excess. An attack that also degrades the references does, and the receiver fails where its network is worse than LMMSE, as at 16-QAM.**

***Index Terms*—Deep receivers, test-time learning, adversarial pilot poisoning, continual learning, physics-informed gating, MIMO detection, physical-layer security.**

I. Introduction

DEEP receivers replace or augment the model-based blocks of a MIMO detector with neural networks trained on data [1]–[4]. Since a network trained in simulation meets a different channel in the field, the strongest of them adapt at test time: online DeepSIC [5] re-trains its per-user networks on the pilots of every coherence block, and Meta-DeepSIC [6] meta-learns an initialisation from the pilots of past blocks so that the per-block fit converges faster. Their adaptation data are the only labelled symbols the receiver sees: a few hundred pilot bits per user per block.

An adversary within range of the base station can add a perturbation to the received pilot part of a block. Wu *et al.* [7] showed that a perturbation shaped through an unrolled copy of the receiver's adaptation step raises the post-adaptation symbol error rate (SER) of online DeepSIC by an order of magnitude, that a perturbation crafted on a surrogate still transfers, and that Meta-DeepSIC, which keeps the poisoned pilots in its support set, does not recover after the attack stops. The receiver cannot verify its adaptation data, because the pilots are the labels; what it can verify is physical consistency: whether its decisions fit a linear channel model, whether the CRC of the decoded data passes, and whether the estimated channel moved as a channel does. And the harm of poisoning lies less in the attacked block than in what the receiver keeps.

TABLE I
Positioning. Attack surface: what the adversary writes to.

Class	Adapts	Memory	Attack surface / defence
LMMSE	no	–	pilot estimate / –
Pilot-spoofing detection [8], [10], [11]	no	–	pilots / detection only
Online DeepSIC [5]	every block	last block	pilots / –
Meta-DeepSIC [6]	every block	pilot buffer	pilots, persistent / –
Code-verified online training [12], [13]	on decoded data	last block	labels / code check, no persistent gate
CRC-Rollback (Section VI-B)	every block	last accepted	pilots / CRC gate, rollback
Filtering and restoring TTA [14]–[17]	every batch	weights, anchor	batch / entropy filter, restore; i.i.d. batches
Robust TTA [22], [23]	every batch	batch statistics	batch / robust statistics
Anchored fine-tuning [19]–[21]	every step	prior weights	data / pull to a prior
Nested, continuum memory [26], [27]	every token	multi-rate	input / –
Citadel-Rx (this paper)	every block	three rates, gated	pilots / physics gates, protected fallback, bounded retention

This paper argues for one principle: *a test-time receiver resists pilot poisoning when it consolidates into persistent memory only what an independent, label-free physical check verifies, and when that check judges every update against the protected memory.* The two parts depend on each other. The simplest receiver with a check and a memory, online DeepSIC with a CRC test and rollback to its last accepted weights, fails: its only reference is LMMSE on the same poisoned pilots, which the attack degrades together with the update, so the first poisoned update passes the relative test, becomes the state the receiver rolls back to, and every later update starts from it (Section VI-B). Citadel-Rx keeps memory at three rates, a fast level that fits every block and two slow levels that change only through a gated, clipped average of the faster states, and its gates compare the fast level with a detector built on the slow memory, which the pilots of the block do not reach. Multi-rate memory is one realisation of the principle; the ablation shows which of its parts are necessary.

A. Rashid is with the Department of Computer Science, Brunel University London, London, U.K. (e-mail: 2546393@brunel.ac.uk).

Code, configuration files, every run record, the statistics and the scripts that regenerate every table, figure and number of this paper are available at https://github.com/abraheemrashid99-sys/Citradel-Rx; one command re-runs the campaign and one rebuilds the paper (Section V).

Table I places the design among the work it draws on. Pilot spoofing is an established physical-layer threat [8], [9], with detectors that flag it [10], [11] but no learning receiver to protect. Decision-directed training on decoded data verified by the code is the closest precedent for using the code check to decide what a receiver learns from [12], [13]; it verifies labels, not what enters a persistent memory. Test-time adaptation in vision filters, anchors or restores what enters the adapted weights [14]–[18], anchors fine-tuning to a prior [19], [20] or averages a teacher [21], and has its own poisoning problem [22] with robust-statistics defences [23]; these methods assume i.i.d. batches and are compared below in receiver form. Byzantine-robust aggregation [24] needs many gradient vectors per step, and certified defences against data poisoning [25] bound the loss of a convex learner; here the learner is a network that sees one correlated block at a time. Nested learning [26] and Titans [27] keep memory at several rates but do not decide what enters the slow rates, and gated fast-to-slow consolidation echoes complementary learning systems [28], [29]. Adversarial examples against deep physical-layer models [30] and the methodology of attacks adapted to a defence [31], [33], [34] shape the evaluation. No published deep receiver combines multi-rate memory, physics-gated consolidation and a retention bound.

The contributions are as follows.

1) **The principle and a receiver that realises it** (Section III): gated, clipped consolidation into two slow levels, a fallback that detects a rejected block with the slow memory, and per-sub-block CRC selection among the fast, slow and LMMSE decisions; the ablation identifies the slow levels, the fallback and a decision-level gate as necessary, and attributes most of the clean-SER gain to the CRC selection.
2) **Retention bounds for the algorithm as implemented** (Section IV): a design bound, with no measured quantity, on the paired displacement of the slowest level under any attacker, and a structural bound for unequal gate verdicts computed from the paired trajectories. A first-version error in the consolidation rule, which broke the convexity the bounds rely on, is corrected.
3) **Relative physics gates with a derived CRC slack** (Section III-D): three label-free tests whose statistics are ratios against a model-based reference on the same block, and a CRC threshold derived from the pass-fraction statistics, which adapts to the block length.
4) **An evaluation built to be hard to pass** (Sections V–VI): tuned baselines, among them CRC-Rollback, anchored, decision-directed and EATA- and CoTTA-style receivers; attacks built against the defence (soft gates, deep unrolling, restarts, expectation over noise, joint pilot and data poisoning of the references, denial of adaptation, optimised slow drift, a rank-limited physical attacker, a grey-box surrogate and a query attacker); settings fixed on calibration seeds by rules stated in advance and results on held-out seeds; randomised attack windows; and the excess SER over a matched clean run with block-bootstrap intervals and seed-level tests.

TABLE II
NOTATION.

U, N	users, base-station antennas
N_p, N_d, L, n	pilot and data symbols per user and block; symbols per CRC sub-block; sub-blocks per user and block
$\theta^{(k)}, K, C_k$	parameters of memory level k; number of levels; period of level k in blocks ($C_1 = 1$)
$\eta_b, \eta^{(k)}$	fast step size of block b; consolidation rate of level $k \geq 2$
a_b, λ, δ	anchor (slow projection); proximal weight; delta-rule decay
γ_b, τ_i, ω	gate verdict of block b; threshold of gate Gi; soft-step width
$F_{\mathrm{nn}}, F_{\mathrm{ref}}, p_{\mathrm{ref}}$	CRC pass fractions of the fast level and the reference; reference symbol-error probability
ρ, z, κ_k	relative CRC slack; statistical margin; consolidation clip of level k
$\boldsymbol{E}_b, \epsilon_b, q$	pilot perturbation; its energy budget; fraction of attacked blocks in the window
$W, n(W)$	window length in blocks; number of level-K consolidations in it
$d_b, D_k(b)$	paired displacement of the fast level and of level k (attacked minus clean run)

Section II gives the system and threat model, Section III the receiver, Section IV the bounds, Section V the protocol, Section VI the results, Section VII their scope, and Section VIII concludes.

II. System Model and Threat Model

A. Uplink and Adaptive Receiver

U single-antenna users transmit QPSK symbols to a base station with N antennas. Time is divided into coherence blocks $b = 1, 2, \ldots$; within block b the channel $\boldsymbol{H}_b \in \mathbb{C}^{N\times U}$ is constant and the received vector at symbol slot n is

$$\boldsymbol{y}_n = f(\boldsymbol{H}_b \boldsymbol{s}_n) + \boldsymbol{w}_n, \qquad \boldsymbol{w}_n \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \tag{1}$$

with f the identity or the per-antenna power-amplifier nonlinearity $f(x) = \tanh(0.5x)$ [6]. Each block carries N_p pilot symbols per user, known to the receiver, and N_d data symbols framed in CRC-protected sub-blocks of L symbols per user. The main channel is the spatial exponential decay (SED) model of [5], $[\boldsymbol{H}_b]_{ij} = c_{j,b}\, e^{-|i-j|}$, with a per-user cosine fading gain $c_{j,b}$; Table V lists the other families. Table II collects the notation.

The receiver is DeepSIC [5]: Q soft-interference-cancellation iterations, each a per-user multilayer perceptron that maps the received vector and the other users' soft decisions to a posterior over the user's symbol. Its parameters are adapted on the pilots of every block, $\theta_b = \Phi(\theta_{b-1}; \boldsymbol{Y}_b)$, by T steps of a first-order optimiser on the pilot cross-entropy, and then detect the data. Online DeepSIC starts every block from θ_{b-1}; Meta-DeepSIC starts from an initialisation meta-learned by MAML [37] on a buffer of the last 20 pilot blocks, refreshed every five blocks [6].

B. Threat Model

The adversary adds $\boldsymbol{E}_b$ to the received pilot matrix $\boldsymbol{Y}_b \in \mathbb{C}^{N\times N_p}$ of an attacked block, with $\|\boldsymbol{E}_b\|_F \leq \epsilon_b$ set by a signal-to-interference ratio (SIR),

$$\epsilon_b = \|\boldsymbol{R}_b\|_F \; 10^{-\mathrm{SIR}/20}, \tag{2}$$

where $\boldsymbol{R}_b = \boldsymbol{Y}_b$ (received energy, noise included, as in [7]) or $\boldsymbol{R}_b = \boldsymbol{S}_b \boldsymbol{H}_b^{\top}$ (users' signal energy only); both are reported. At SIR 0 dB the attacker's received pilot energy equals that of all U users together (plus the noise, with the first budget).

TABLE III
ATTACKER KNOWLEDGE.

White-box	victim's current weights, adaptation map, gate thresholds and running statistics; differentiates through them
Grey-box	the defence and its settings, not the weights: a Citadel-Rx surrogate trained on the pilots it observes
Transfer	an online DeepSIC surrogate trained on the pilots it observes
Query	no model: the victim's per-sub-block ACK/NACK feedback, searched with natural-evolution strategies [35], [36]

As an order of magnitude, with a common path-loss exponent α, users at a common distance and equal transmit powers, the attacker must be $U^{1/\alpha}$ times closer to the base station than the users: about 1.9 times for $U = 4$ and the line-of-sight urban-micro exponent $\alpha = 2.1$ below the breakpoint distance [39]. The attacker is synchronised to the frame and knows the pilot sequence, which is standardised. The attackers are:

- *A1, availability* [7]: projected gradient ascent [38] on the post-adaptation data loss through an unrolled copy of the adaptation map.
- *A1 built against the defence*: deeper unrolling (10, 50 steps), more steps and random restarts, the soft-gate objective (the data loss of the fast level, which the receiver emits if it accepts the block, times a differentiable surrogate of the probability that its gates accept, plus a reward for acceptance; the surrogate uses the defence's exact thresholds and running statistics with the hard steps replaced by soft ones, as in BPDA [31]), and an expectation over noise draws (EOT [32]).
- *A4, reference degradation*: a fraction α of the energy on the pilots and $1-\alpha$ on the data, optimised jointly so that the model-based references the gates compare against degrade with the fast level.
- *Denial of adaptation*: a fixed channel-estimate spoof $\boldsymbol{E}_b = \boldsymbol{S}_b\boldsymbol{\Delta}^\top$, meant to make the gates reject every block, combined with a real abrupt channel change.
- *A3*, optimised slow drift*: small perturbations over hundreds of blocks that pull the receiver toward a spoofed channel while keeping its blocks accepted.
- *Physical attacker*: $\boldsymbol{E}_b = \boldsymbol{G}\boldsymbol{X}_b$ with $M \in \{1, 2\}$ attacker antennas, a channel $\boldsymbol{G}$ from the same family, and an error on the attacker's estimate of $\boldsymbol{G}$.
- *Data jamming*: the same energy on the data part, a control that separates persistent poisoning from plain interference.

Table III states what each attacker knows.

C. Metrics

The SER is the fraction of symbol slots in which at least one user is wrong [5]–[7]; the CRC is checked per user and sub-block, and the CRC gate uses the per-user pass fractions. The primary metric is the *excess SER*: the SER of an attacked run minus that of the matched clean run (same configuration, seed and channel draws, attack disabled, same machine), averaged over the attack window. Two runs of a pair are identical up to the first attacked block, so the excess isolates the attack from the difficulty of the blocks it hits. The gate is characterised by the rejection rate of poisoned blocks and the false rejection rate of clean blocks, each with its count and a Wilson interval

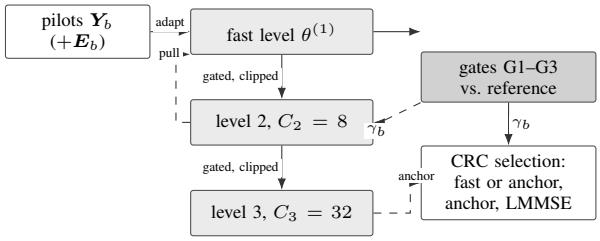


Fig. 1. Citadel-Rx. The fast level fits every block; the gates compare its decisions with a model-based reference on the same block; levels 2 and 3 receive gated, clipped averages of the faster states; a rejected block is reverted and detected with the anchor; every CRC sub-block keeps the first passing detection.

[45]. The paired displacement $D_k(b) = \left\|\theta^{(k)}_{\mathrm{adv},b} - \theta^{(k)}_{\mathrm{clean},b}\right\|$ measures what the attack left in level k.

III. CITADEL-RX

The receiver splits adaptation by time-scale: a fast level fits every block; slow levels change only through what the gates accept (Fig. 1, Algorithm 1).

A. Levels and the Fast Learner

The adaptable parameters $\theta \in \mathbb{R}^d$ (all DeepSIC weights, $d = 67\,632$ for 4×4 QPSK) are kept at $K = 3$ levels with periods $C = (1, 8, 32)$ blocks. The fast level is adapted on every block by $T = 300$ steps of

$$\theta \leftarrow \theta - \eta_b \delta\, \theta\, \boldsymbol{\phi}\boldsymbol{\phi}^\top \quad \text{(delta-rule decay, per weight matrix)}, \tag{3}$$

$$\theta \leftarrow \theta - \eta_b \lambda(\theta - a_b) \quad \text{(proximal pull to the anchor)}, \tag{4}$$

$$\theta \leftarrow \mathrm{Adam}\big(\theta, \nabla_\theta \ell_b(\theta)\big), \tag{5}$$

with ℓ_b the pilot cross-entropy of the stage being adapted (stages are adapted sequentially [5]), $\boldsymbol{\phi}$ the normalised mean input of the stage, and $a_b = \frac{1}{2}\theta^{(2)}_{b-1} + \frac{1}{2}\theta^{(3)}_{b-1}$ the anchor. The step size follows the receiver's surprise $S_b = \|\nabla \ell_b(\theta_{b-1})\|^2$ relative to its moving average $\bar{S}_b$: $\eta_b = \eta_{\max}\, 2r_b/(1+r_b)$ with $r_b = \mathrm{clip}(S_b/\bar{S}_b, \frac{1}{8}, 8)$, so the rate is bounded by $2\eta_{\max}$.

B. Gated, Clipped Consolidation

At the end of every period of level $k{+}1$ the level receives the gated mean of the level-k states buffered since its last update, each weighted by the verdict $\gamma_j \in [0, 1]$ of its block (or the mean verdict of its period):

$$\theta^{(k+1)} \leftarrow \theta^{(k+1)} + \Pi_{\kappa_{k+1}}\Big(s\big(m - \theta^{(k+1)}\big)\Big),$$
$$m = \frac{\sum_j \gamma_j \theta^{(k)}_j}{\sum_j \gamma_j}, \qquad s = \eta^{(k+1)} \min\big(1, \textstyle\sum_j \gamma_j\big), \tag{6}$$
$$\Pi_\kappa(\boldsymbol{v}) = \boldsymbol{v} \min\{1, \kappa/\|\boldsymbol{v}\|\},$$

with $\eta^{(2)} = 1$, $\eta^{(3)} = \frac{1}{2}$, and no update when $\sum_j \gamma_j = 0$. The clip $\kappa = (8.5, 4.5)$ lies at or just above the largest clean consolidation increments of levels 2 and 3 on the calibration runs (8.2 and 4.5), so it rarely binds on clean runs (in 5/150 consolidations of the clean twins at the headline cell) and limits what any attacked period can move. The first version of

Algorithm 1 Citadel-Rx, one block b

Require: pilots $\boldsymbol{Y}_b$ (possibly poisoned), data $\boldsymbol{Y}_b^d$; levels $\theta^{(1..K)}$; buffers; running statistics
$a_b \leftarrow \sum_{k\geq 2} w_k \theta^{(k)}$; $\eta_b \leftarrow \eta_{\max} 2r_b/(1+r_b)$
reference: $\hat{\boldsymbol{H}}_b$ (LS, AR(1) tracker); $\hat{\boldsymbol{s}}_{\mathrm{mm}}$ (LMMSE); $\hat{\boldsymbol{s}}_a$ (anchor)
$\theta \leftarrow \theta^{(1)}$; T steps of (3)–(5) on $\boldsymbol{Y}_b$; $\hat{\boldsymbol{s}}_{\mathrm{nn}} \leftarrow \mathrm{detect}(\theta, \boldsymbol{Y}_b^d)$
$\gamma_b \leftarrow \min(\gamma_1, \gamma_2, \gamma_3)$: G1 residual ratio, G2 CRC ratio $F_{\mathrm{nn}}/F_{\mathrm{ref}}$, G3 innovation
if $\gamma_b < \frac{1}{2}$ **then**
    $\theta \leftarrow \theta^{(1)}$ (revert); $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_a$ (fallback)
else
    $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_{\mathrm{nn}}$; update the running statistics
end if
per user and sub-block: $\hat{\boldsymbol{s}} \leftarrow$ first CRC-passing of $(\hat{\boldsymbol{s}}, \hat{\boldsymbol{s}}_a, \hat{\boldsymbol{s}}_{\mathrm{mm}})$
$\theta^{(1)} \leftarrow \theta$; buffer $(\gamma_b, \theta^{(1)})$; for every k with $(b+1) \bmod C_{k+1} = 0$: (6)
return $\hat{\boldsymbol{s}}$

the receiver divided the gated sum by $\max(1, \sum_j \gamma_j)$; when most blocks of a period are rejected, which is the case under attack, this pulls the level toward the origin instead of leaving it nearly unchanged, and the update is then not the convex combination that the bounds of Section IV require. Every result of this revision uses (6).

C. Rejection, Fallback and CRC Selection

If $\gamma_b < \frac{1}{2}$ the block is rejected: the fast update is reverted, the reverted state enters the buffer with weight γ_b, and the data of the block are detected with the anchor, which the attacker has not reached. Each CRC sub-block of each user then keeps the first of three detections that passes its CRC: the gate's choice (fast level or anchor), the anchor, and LMMSE on the least-squares channel estimate. All three are computed for the gates anyway; when none passes, the gate's choice is kept. Three optional states answer specific attacks and are evaluated with them, with mixed results (Sections VI-D and VI-E): *abstain*, the block is rejected and not buffered when the reference itself has degraded (its pass fraction or residual is $z = 4$ standard deviations from its running level on accepted blocks, with a binomial floor on the variance); *streak recovery*, after five consecutive rejections a re-fit from the anchor on CRC-verified data, kept if it verifies better than the reference; and a *drift guard*, a pull of level K toward the pre-trained weights by 5% per consolidation.

D. Relative Physics Gates

Every statistic is a ratio against a reference computed on the same block without labels: the LMMSE detector on the least-squares pilot estimate $\hat{\boldsymbol{H}}_b$ and the anchor detector, the better of the two. Ratios are normalised by their moving averages over accepted blocks and passed through a soft step of width $\omega = 0.1\tau$, tightened by up to 30% when $r_b > 1$.

G1, model residual. With $r(\hat{\boldsymbol{s}}) = \frac{1}{N_d}\sum_n \left\| \boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \hat{\boldsymbol{s}}_n \right\|^2$, the statistic is $r(\hat{\boldsymbol{s}}_{\mathrm{nn}})/r(\hat{\boldsymbol{s}}_{\mathrm{ref}})$ against $\tau_1 = 1.25$.

G2, CRC pass rate. The pass fraction F_{nn} of the fast level over F_{ref} against the threshold τ_2 of the design rule below. The gate is tied to the availability attacker's own objective: an attack that raises the SER lowers the pass rate.

G3, channel prior. The innovation energy of the LS estimate against its AR(1) prediction, relative to its running level and floored by the $\chi^2_{0.999}$ quantile, against $\tau_3 = 4$.

Table IV gives the provenance of every setting.

TABLE IV
SETTINGS AND THEIR PROVENANCE. CALIBRATION: SEEDS 0–2 BY THE RULES OF SECTION V; FIRST VERSION: SET DURING THE DEVELOPMENT OF THE FIRST VERSION ON SEEDS 0–4. NO SETTING WAS CHOSEN ON THE TEST SEEDS 10–19.

Setting	Value; provenance
Q, MLP width, T, $\eta_{\max}$, Adam	3, 64·order (= 256 for QPSK), 300, $5 \cdot 10^{-3}$; DeepSIC reference [5]
K, C	3, $(1, 8, 32)$; first version; ablation in Section VI-G
$\eta^{(2)}, \eta^{(3)}$, λ, δ	1, $\frac{1}{2}$, 0.5, 0.1; first version; sensitivity in Section VI-I
τ_1, τ_3, ω, tightening	1.25, 4, 0.1τ, 30%; first version; sensitivity
CRC slack	relative $\rho = 1$ with floor $2 \cdot 10^{-3}$, or derived $z = 5$; mode chosen by calibration (relative slack)
κ	$(8.5, 4.5)$; at the largest clean consolidation increments of the calibration runs
Consolidation rule, CRC selection	(6); correctness fix; CRC selection kept by calibration
Baseline epochs, learning rate, weight decay	tuned on calibration seeds (Section V)
Attacker	PGD 20 steps of 0.25ϵ, 2 unrolled steps [7]; stronger variants in Section VI-C

IV. RETENTION BOUNDS

The bounds are stated for the algorithm as implemented, for a pair of runs on the same channel, noise and seed, with and without an attack that starts at block b_0 (the runs are identical before b_0). Let $n(W)$ be the number of level-K consolidations in $[b_0, b_0 + W)$, so $n(W) \leq \lceil W/C_K \rceil$; quantities of the clean run are primed.

Lemma 1 (convex consolidation)**.** *Without the clip,* (6) *writes the new level-$(k{+}1)$ state as the convex combination $(1-s)\theta^{(k+1)} + s\,m$ of its old state and the buffered states, with $s \leq \eta^{(k+1)}$ and m in their convex hull. With the clip the step is $\tilde{s} = s \min\{1, \kappa_{k+1}/\left\|s(m - \theta^{(k+1)})\right\|\} \leq s$, and the statement holds with $\tilde{s}$.*

Theorem 1 (retention)**.** *(i)* Design bound. *With the clip, for any perturbations and any gate verdicts,*

$$D_K(b_0 + W) \leq 2\kappa_K\, n(W). \tag{7}$$

(ii) Structural bound. *Let $\Delta(W)$ be the largest paired displacement of the buffered level-$(K{-}1)$ states in the window and, at the i-th consolidation, $\bar{w}_i, \bar{w}'_i$ the normalised weights, $\tilde{s}_i, \tilde{s}'_i$ the steps taken, m'_i the clean gated mean and r'_i the radius of the clean buffer around it. Then*

$$D_K(b_0 + W) \leq \big(\Delta(W) + \Omega(W)\big)\big(1 - (1-\eta^{(K)})^{n(W)}\big) + \sum_{i \leq n(W)} |\tilde{s}_i - \tilde{s}'_i| \left\| \theta'^{(K)}_i - m'_i \right\|, \tag{8}$$

with $\Omega(W) = \max_i \|\bar{w}_i - \bar{w}'_i\|_1\, r'_i$. With equal verdicts $\Omega = 0$, the sum vanishes, and $\Delta(W)$ is at most the largest paired fast-level displacement.

Proof. (i) Level K changes only at consolidations, and in each run by at most κ_K; the drift guard, when used, multiplies the paired difference by 0.95. (ii) At consolidation i, $\theta^{(K)}_{\mathrm{new}} - \theta'^{(K)}_{\mathrm{new}} = (1-\tilde{s}_i)(\theta^{(K)} - \theta'^{(K)}) + \tilde{s}_i(m_i - m'_i) + (\tilde{s}_i - \tilde{s}'_i)(m'_i - \theta'^{(K)})$. Since $\sum_j \bar{w}_{ij} = \sum_j \bar{w}'_{ij} = 1$, $m_i - m'_i = \sum_j \bar{w}_{ij}(\theta_j - \theta'_j) + \sum_j (\bar{w}_{ij} - \bar{w}'_{ij})(\theta'_j - m'_i)$, whose norm is at most $\Delta + \Omega$.

Iterating $D \leftarrow (1-\tilde{s}_i)D + \tilde{s}_i(\Delta+\Omega) + e_i$ from $D=0$ gives $D \le (\Delta+\Omega)(1-\prod_i(1-\tilde{s}_i)) + \sum_i e_i$, and $\tilde{s}_i \le \eta^{(K)}$. □

Bound (7) needs no measured quantity and holds for every attacker of this paper, including those that pass the gates; its price is that a legitimate change of the channel moves the slowest level by at most κ_K per period. Bound (8) is evaluated on the paired trajectories in Section VI-E; it is tighter when the verdicts agree, and it is the form in which the gate enters: rejected blocks are reverted and enter with weight γ_b.

Remark 1 (contraction). If the block map satisfied $\|\Phi(\theta;\boldsymbol{Y}+\boldsymbol{E},a) - \Phi(\theta;\boldsymbol{Y},a)\| \le G_\Phi \|\boldsymbol{E}\|$ and were jointly L-Lipschitz in (θ,a) with $L<1$, the largest fast displacement after n_W accepted attacked blocks would satisfy $\Delta_f \le G_\Phi\epsilon(1-L^{n_W})/(1-L)$. The sufficient condition is $L = L_\theta + L_a < 1$. Finite-difference probes of the fast learner, which the revision does not change, give $L_\theta + L_a$ between 1.08 and 1.39 for proximal weights λ from 0.5 to 4; probes can only underestimate a Lipschitz constant, so the condition fails and this branch cannot be used for the default receiver. The regret bound of the first version, which relied on it, is withdrawn; a last-layer-adaptable variant, for which the loss is convex, is evaluated in Section VI-H without a regret claim.

Design rule for the CRC gate. With i.i.d. symbol errors of probability p in a sub-block of L symbols and an r-bit CRC whose syndrome is uniform on error, $P(\text{pass} \mid p) = (1-p)^L + 2^{-r}(1-(1-p)^L)$. Inverting at F_{ref} gives p_{ref}, and the rule "reject if $F_{\text{nn}}/F_{\text{ref}} < \tau_2$" with $\tau_2 = \big((1-p_{\text{ref}}-\delta)/(1-p_{\text{ref}})\big)^L$ tolerates an excess error probability δ over the reference. The relative slack sets $\delta = \max\{2\cdot10^{-3}, \rho\, p_{\text{ref}}\}$; the derived slack removes the floor and takes $\tau_2 = \min\{1 - z\sigma, \tau_2(\rho p_{\text{ref}})\}$, with σ the standard deviation of the normalised pass ratio on accepted blocks, floored by its binomial value $\sqrt{F(1-F)/n}/F$, so the threshold follows the block length. For n independent sub-blocks, $P(|\hat{F}-F| \ge t) \le 2e^{-2nt^2}$ [43]. The formula is standard; its use here is to replace a tuned threshold by a target excess error.

Model mismatch and G1. Writing $\boldsymbol{y}_n = \hat{\boldsymbol{H}}\boldsymbol{s}_n + \boldsymbol{\nu}_n + \boldsymbol{m}_n$ with $\boldsymbol{m}_n$ the mismatch, the residual of a decision with error $\boldsymbol{e}_n$ is $\text{mean}\left\|\hat{\boldsymbol{H}}\boldsymbol{e}_n\right\|^2 + \text{mean}\,\|\boldsymbol{\nu}_n + \boldsymbol{m}_n\|^2$ plus a cross term that vanishes if decision errors are uncorrelated with noise and mismatch. This is an assumption, not a fact: errors occur when the noise is large. On clean blocks of the tanh channel the normalised G1 statistic has mean 1.000, 99th percentile 1.050 and maximum 1.080 over 285 blocks, and never reaches $\tau_1 = 1.25$.

V. Experimental Protocol

Receivers. LMMSE on the least-squares pilot estimate, and a Huber-robust version [42]; online DeepSIC [5]; Meta-DeepSIC [6]; *CRC-Rollback*, online DeepSIC with the CRC gate and rollback to the last accepted weights, and the same with all three gates; online DeepSIC with the CRC selection of Citadel-Rx, which separates the selection gain from the memory; L2-SP [19], EWC with the Fisher information of the pre-training data [20] and an EMA teacher [21]; CRC-verified decision-directed training, which adds the CRC-passing data symbols to the pilots [12], [13]; an EATA-style receiver, entropy filter and Fisher anchor [14], and a CoTTA-style receiver, teacher and stochastic restore [15]; a drift-detection receiver; a median-of-means gradient receiver after [23]; and a single-memory receiver with the fast learner of Citadel-Rx and no slow levels [27]. All DeepSIC receivers share the architecture and the pre-training.

TABLE V
Simulation setup.

System	$U \times N = 4\times4$, QPSK, SNR 10 dB; 4×16; 16-QAM at 20 dB (MLP width kept at 256)
Block	$N_p = 100$ pilot and $N_d = 25\,000$ data symbols per user; $N_d \in \{1000, 5000\}$, $N_p \in \{32, 100\}$ in Section VI-H
CRC	CRC-8 of [40], $D^8+D^7+D^4+D^3+D+1$; $L=32$; $n=781$ sub-blocks per user and block
Channel	SED with per-user cosine fading (periods 51, 39, 33, 21 blocks; depth 0.8) [5]; COST2100 traces of [6], [41] in calibration
Run	100 blocks (400 for A3*); 40-block attack window starting at a block drawn uniformly from 10–50 per seed
Seeds	calibration 0–2; test 10–19 (headline cell), 10–12 (other headline cells), 10–11 (other experiments), 10 (sensitivity)
Attacker	PGD 20 steps of 0.25ϵ through 2 unrolled steps [7], [38], unless stated

Calibration and tuning, fixed in advance. Online DeepSIC and Meta-DeepSIC are tuned over 300, 100 and 30 adaptation epochs, learning rates 10^{-3} and $5\cdot10^{-3}$ and weight decay 0 and 10^{-4} on clean runs of the calibration seeds; the lowest mean clean SER wins (online DeepSIC: 30 epochs, learning rate 0.001, weight decay 0; Meta-DeepSIC: 300, 0.001, 0), and every receiver built on online DeepSIC uses its setting. Citadel-Rx has three candidates, relative or derived CRC slack and, with the relative slack, a fast level restarted from the anchor every block; the rule, fixed before the runs, is the lowest mean of the clean SER, the A1 window SER at SIR 0 dB, and the A1 window SER on COST2100, among candidates with at most 2% clean false rejections (Table VI). The chosen settings are written to a file by the rule and read by every test run.

Statistics. Within a run, consecutive blocks are correlated (decorrelation lengths of 6 blocks for the excess SER at the headline cell), so intervals are 95% moving-block bootstrap intervals [44] that resample seeds and then runs of twice the decorrelation length (at least five blocks) within each seed. Tests are at the seed level: an exact two-sided Wilcoxon signed-rank test on the per-seed means [46], whose smallest attainable p with ten seeds is 0.002, Holm-corrected over the four baselines of a column [47]. Rates carry their counts and Wilson intervals.

Reproducibility. The campaign ran on an Apple M4 laptop (ten cores, eight runs at a time) and a 4-core cloud machine (four at a time), every run single-threaded, with disjoint sets of runs; times per block are wall-clock times of such runs and compare receivers only roughly. Every record stores its configuration, seed, code version and timings; an attacked run and its clean twin always ran on the same machine. Pre-training is cached by configuration, seed, code and random-number state, which is exact (the repository's tests compare cached and uncached runs). One command re-runs the campaign and one recomputes every number and table of this paper.

TABLE VI
CALIBRATION OF CITADEL-RX ON SEEDS 0–2 (SER). SCORE: MEAN OF THE CLEAN, A1 AND COST2100 A1 COLUMNS.

Candidate	Clean	A1 window	COST2100 A1	False rej.	Score	
Relative slack	0.0151	0.0245	$2.8 \cdot 10^{-4}$	0.000	0.0133	chosen
Derived slack	0.0150	0.0272	$2.3 \cdot 10^{-4}$	0.000	0.0141	
Relative slack, anchor start	0.0168	0.0318	$2.2 \cdot 10^{-4}$	0.000	0.0163	

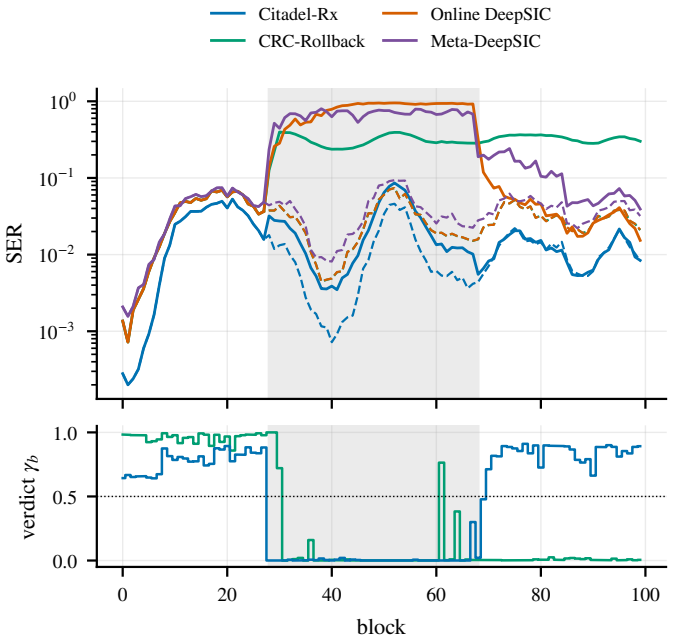


Fig. 2. Test seed 10, white-box A1 at SIR 0 dB in the shaded window. Top: SER per block of the attacked runs (solid) and their clean twins (dashed). Bottom: gate verdict γ_b of CRC-Rollback and Citadel-Rx; a block is rejected below $\frac{1}{2}$.

VI. RESULTS

A. Headline Comparison

Table VII contains the main result. Under white-box A1 at SIR 0 dB the attack raises the SER of Citadel-Rx in the attack window by $+0.0188$ over its clean twin (95% interval [0.0142, 0.0246]; seed means from $+0.0130$ to $+0.0279$), and that of tuned online DeepSIC by $+0.768$. The excess of Citadel-Rx is small but not zero: it is positive on 10 of 10 seeds (exact Wilcoxon $p = 0.002$). The gates reject 397/400 poisoned blocks and 1/950 clean blocks of the clean twins. The ordering is the same at SIR -10 and $+10$ dB and under transfer (three seeds each). Over the 30 blocks after the window the excess of Citadel-Rx is $+3.8 \cdot 10^{-4}$; online DeepSIC keeps $+0.0189$, Meta-DeepSIC $+0.0882$, whose meta-training buffer keeps poisoned pilots for 20 blocks after the attack, and CRC-Rollback $+0.346$. Fig. 2 shows one seed.

Paired seed by seed (Table VIII), Citadel-Rx has the lower window SER against each of the four baselines on every seed (Holm-corrected $p = 0.008$, the smallest attainable with ten seeds and four tests), and also the lower clean SER: -52% relative to tuned online DeepSIC. Most of the clean gain, 70%, comes from the per-sub-block CRC selection rather than from the memory: online DeepSIC with the same selection has a clean SER of 0.0209, against 0.0330 without it and 0.0159 for Citadel-Rx, and the selection does not protect online DeepSIC (excess $+0.765$). The jamming column is a control, not an attack these receivers are built to resist: at the same SIR, jamming spreads its power over the data symbols, 250 times the energy of the pilot attack, and raises the SER of every receiver by about the same amount ($+0.816$ to $+0.837$); Citadel-Rx rejects 4/120 of the jammed blocks, whose pilots are clean.

B. Why Rollback Is Not Enough

CRC-Rollback has the CRC gate and the rollback of Citadel-Rx but one memory, and it fails in every column of Table VII. The records show why (Fig. 2). Its only reference is LMMSE on the least-squares estimate from the same pilots, and the attack degrades that estimate: on attacked blocks the reference passes a fraction 0.19 of the CRCs, against 0.71 on the same blocks of the clean twin. A poisoned fast level that is merely less bad than this reference passes the relative test: the first attacked block was accepted on 10 of 10 seeds (41/400 attacked blocks accepted in all). The accepted state then becomes the rollback state; every later update starts from it, fails the test and is reverted to it. After the attack the receiver rejects a fraction 1.00 of the clean blocks, and over the last ten blocks of the run its excess SER is still $+0.351$. Rollback with all three gates fails in the same way (two seeds: 8/80 attacked blocks accepted, excess $+0.241$ at the end of the run), and so does Citadel-Rx without its slow levels (Section VI-G).

Citadel-Rx compares the fast level with the better of LMMSE and the anchor detector, which detects the data with the slow memory and does not use the pilots of the block. Under attack this reference passes 0.76 of the CRCs (0.78 on the clean twin) while the poisoned fast level passes 0.21; 3/400 attacked blocks were accepted, on 3 of the 10 seeds, and the excess over the last ten blocks is $-8.4 \cdot 10^{-5}$. The two parts of the principle depend on each other: the gate protects the memory, and it can only do so because it judges against the memory.

C. Attacks Built Against the Defence

Every attack of Table IX was built with full knowledge of the defence or of its feedback and run on test seeds 10–11. Against Citadel-Rx at SIR 0 dB the 11 attacks give excess SERs between $+0.0125$ and $+0.0148$, against $+0.0143$ for plain A1 on the same seeds, with rejection rates between 0.925 and 1: deeper unrolling, more steps and restarts, the soft-gate objective with the exact thresholds and running statistics, expectation over noise, a Citadel-Rx surrogate and a search on the NACK feedback do not get poisoned blocks accepted. At SIR 10 dB the excess is $+0.0064$ to $+0.0084$. The same attacks raise the SER of online DeepSIC by $+0.228$ (rank-1 physical attacker) to $+0.863$. With two seeds, differences of a few 10^{-3} between attacks lie within the intervals; what the table rules out is an attack that brings Citadel-Rx near the excess of the undefended receivers.

TABLE VII
EXCESS SER (ATTACKED MINUS MATCHED CLEAN RUN, ATTACK WINDOW) AT THE HEADLINE CELLS, TEST SEEDS; 95% MOVING-BLOCK INTERVAL AT WHITE-BOX SIR 0 DB (10 SEEDS; OTHER COLUMNS 3 SEEDS). CLEAN: SER OF THE CLEAN TWINS OVER BLOCKS 5–99. REJECTION AND FALSE REJECTION OF THE CITADEL-RX GATES AT WHITE-BOX SIR 0 DB. SCRIPT: REVISION_NUMBERS.PY.

Receiver	Clean	WB −10 dB	WB 0 dB	WB +10 dB	Transfer 0 dB	Jamming 0 dB	Rej.	False rej.
Citadel-Rx	0.0159	+0.0166	+0.0188 [0.0142, 0.0246]	+0.0088	+0.0147	+0.837	0.993	0.001
Online DeepSIC	0.0330	+0.869	+0.768 [0.715, 0.815]	+0.296	+0.382	+0.822	–	–
Meta-DeepSIC	0.0393	+0.897	+0.651 [0.630, 0.670]	+0.195	+0.194	+0.816	–	–
CRC-Rollback	0.0330	+0.432	+0.271 [0.234, 0.307]	+0.0642	+0.264	+0.824	0.897	0.000
Online + CRC selection	0.0209	+0.880	+0.765 [0.711, 0.814]	+0.181	+0.376	+0.833	–	–

TABLE VIII
PAIRED SEED-LEVEL COMPARISON, CITADEL-RX MINUS BASELINE: ATTACK-WINDOW SER AT WHITE-BOX SIR 0 DB AND CLEAN SER (RELATIVE CHANGE), EXACT WILCOXON p, HOLM-CORRECTED OVER THE FOUR BASELINES.

Baseline	Seeds	Δ window SER	p	Δ clean SER	p
Online DeepSIC	10	−0.765	0.008	−0.0171 (-52%)	0.008
Meta-DeepSIC	10	−0.656	0.008	−0.0234 (-60%)	0.008
CRC-Rollback	10	−0.269	0.008	−0.0171 (-52%)	0.008
Online + CRC selection	10	−0.750	0.008	−0.0051 (-24%)	0.008

TABLE IX
ATTACKS BUILT AGAINST THE DEFENCE, SIR 0 DB UNLESS STATED, TEST SEEDS 10–11. [a]SOFT GATES, UNROLL 10, PGD 50, 2 RESTARTS, EOT OVER 2 NOISE DRAWS; [b]THE SAME WITHOUT SOFT GATES, WHICH ONLINE DEEPSIC DOES NOT HAVE.

Attack	Receiver	See-ds	Excess SER [95% CI]	Window SER	Rej.
Unroll 10	Citadel-Rx	2	+0.0137 [0.0081, 0.0207]	0.0250	1.000
Unroll 50	Citadel-Rx	2	+0.0138 [0.0082, 0.0208]	0.0251	1.000
PGD 100, 5 restarts	Citadel-Rx	2	+0.0140 [0.0083, 0.0212]	0.0254	1.000
Soft gates (BPDA), unroll 10	Citadel-Rx	2	+0.0136 [0.0079, 0.0206]	0.0249	1.000
Combined[a]	Citadel-Rx	2	+0.0139 [0.0081, 0.0212]	0.0252	1.000
Combined[b]	Online DeepSIC	2	+0.863 [0.790, 0.922]	0.889	–
Unroll 10, SIR 10 dB	Citadel-Rx	2	+0.0084 [0.0041, 0.0139]	0.0197	0.938
Soft gates, SIR 10 dB	Citadel-Rx	2	+0.0064 [0.0030, 0.0107]	0.0177	1.000
PGD 100 x 5, SIR 10 dB	Citadel-Rx	2	+0.0071 [0.0035, 0.0115]	0.0184	1.000
Rank-1 physical	Citadel-Rx	2	+0.0125 [0.0073, 0.0193]	0.0238	0.925
Rank-1 physical	Online DeepSIC	2	+0.228 [0.135, 0.405]	0.254	–
Rank-2 physical	Citadel-Rx	2	+0.0143 [0.0084, 0.0216]	0.0256	1.000
Rank-2 physical	Online DeepSIC	2	+0.586 [0.496, 0.680]	0.612	–
Rank-2, CSI error 0.1	Citadel-Rx	2	+0.0143 [0.0084, 0.0216]	0.0256	1.000
Rank-2, CSI error 0.1	Online DeepSIC	2	+0.572 [0.487, 0.662]	0.599	–
Grey-box (Citadel-Rx surrogate)	Citadel-Rx	2	+0.0148 [0.0085, 0.0224]	0.0261	1.000
Query (NACK feedback)	Citadel-Rx	2	+0.0136 [0.0081, 0.0206]	0.0249	1.000
Query (NACK feedback)	Online DeepSIC	2	+0.527 [0.401, 0.643]	0.553	–
Signal-referenced budget	Citadel-Rx	2	+0.0140 [0.0082, 0.0212]	0.0253	0.988
Signal-referenced budget	Online DeepSIC	2	+0.725 [0.593, 0.848]	0.752	–

When the gates reject nearly every attacked block, the excess no longer depends on the attack: it is the price of rejecting. The window is detected with the anchor, which has not adapted to the block, and the LMMSE alternative of the CRC selection uses the poisoned channel estimate (its pass fraction falls from 0.71 on the clean twin to 0.20 on attacked blocks). On the poisoned blocks at the headline cell, the verdict of G2 on its own is below one half for 99% of them, that of G1 for 88% and that of G3 for 14%: the decision-level gates G1 and G2 do the work, and G3 flags few poisoned blocks at its threshold.

TABLE X
JOINT PILOT AND DATA ATTACK ON THE REFERENCES (A4), WHITE-BOX SIR 0 DB, 6 PGD STEPS OVER ALL DATA SUB-BLOCKS, TEST SEEDS 10–11, WITH AND WITHOUT THE ABSTAIN STATE (WHICH REJECTS EVERY BLOCK OF ITS CLEAN RUNS AS WELL, SEE TEXT); JAMMING CONTROLS WITH THE DATA POWER OF A4.

Attack	State	See-ds	Excess SER [95% CI]	Window SER	Rej.
$\alpha = 0.25$	–	2	+0.0420 [0.0216, 0.0652]	0.0533	0.575
$\alpha = 0.25$	abstain	2	+0.0233 [0.0112, 0.0383]	0.0452	1.000
jamming	0.75 of power	2	+0.787	–	–
$\alpha = 0.75$	–	2	+0.0322 [0.0143, 0.0547]	0.0435	0.988
$\alpha = 0.75$	abstain	2	+0.0250 [0.0113, 0.0413]	0.0468	1.000
jamming	0.25 of power	2	+0.476	–	–

TABLE XI
ABRUPT CHANNEL REDRAW AT BLOCK 50, WITHOUT AND WITH THE DENIAL-OF-ADAPTATION SPOOF ON BLOCKS 40–89 (SIR 10 DB), AND A 6 DB SNR DROP AT BLOCK 50; TEST SEEDS 10–11. SER AFTER THE CHANGE (NO ATTACK), FALSE REJECTION AFTER IT, EXCESS SER OF THE SPOOF IN AND AFTER ITS WINDOW, SER AFTER THE SNR DROP.

Receiver	See-ds	SER after redraw	False rej.	Spoof excess	After spoof	SER after drop
Citadel-Rx	2	0.283	0.230	+0.106	+0.0101	0.456
+ streak recovery	2	0.283	0.230	+0.106	+0.0096	0.456
Online DeepSIC	2	0.239	–	+0.198	+0.0674	0.468
LMMSE	2	0.444	–	+0.184	0	0.479

D. Attacks on the References and on Adaptation

A4 splits the attacker's power between the pilots and the data so that the references the gates compare against degrade together with the fast level (Table X). It is the most effective attack on Citadel-Rx in this paper. With a quarter of the power on the pilots, the reference passes 0.71 of the CRCs on attacked blocks (0.82 on the clean twin), close enough to the poisoned fast level (0.46) that only 46/80 attacked blocks are rejected, and the excess SER is $+0.0420$, against $+0.0143$ for A1 on the same seeds; with three quarters on the pilots it is $+0.0322$. An attacker that reaches the data does far more harm by jamming it: the same data power as noise raises the SER by $+0.787$ and $+0.476$ for $\alpha = 0.25$ and 0.75. The abstain state, meant to answer A4, does not work as implemented. Its running statistics are updated only on accepted blocks and its

TABLE XII
RETENTION OF THE SLOWEST LEVEL: PAIRED DISPLACEMENT D_K (LARGEST IN THE WINDOW, MEAN OVER SEEDS) AGAINST THE STRUCTURAL BOUND (8) AND THE DESIGN BOUND (7). A1: PAIRED RUNS WITH LEVEL SNAPSHOTS, FIXED WINDOW 30–69, SEEDS 10–12. A3*: BLOCKS 30–329 OF 400, ENERGY RELATIVE TO THE PILOTS, SEEDS 10–11.

Attack	Seeds	Excess SER [95% CI]	Rej.	D_K	Struct.	Design
A1, SIR 0 dB	3	+0.0136 [0.0086, 0.0198]	0.992	4.11	15.4	18.0
A1, SIR 10 dB	3	+0.0072 [0.0040, 0.0111]	0.958	3.78	13.5	18.0

variance floor is binomial over thousands of CRC checks, so the fading of the channel alone, which lowers the reference pass fraction from 0.987 to 0.963, trips it at block 4–5; from then on it rejects every block, clean or attacked (191/200 blocks of its clean twins rejected, clean SER 0.0311). Its rejection of every A4 block and its lower excess in Table X are those of a receiver that no longer adapts in either run, not a defence. A usable abstain state needs a reference model that follows legitimate change; it is not evaluated here.

An abrupt channel redraw at block 50 is the legitimate change the gates must not block (Table XI). After it Citadel-Rx rejects a fraction 0.230 of the clean blocks and its SER over the remaining blocks is 0.283, against 0.239 for online DeepSIC and 0.444 for LMMSE: the gates slow the re-adaptation, a cost of the defence. After a 6 dB SNR drop, which the receivers are not told about, all three are close (0.456, 0.468, 0.479). The denial spoof did not achieve its aim of tripping the gates on every block: they rejected fractions 0.04 and 0.32 of the spoofed blocks on the two seeds. It acted instead as an accepted perturbation during the change, raising the SER in its window by +0.106 for Citadel-Rx, +0.198 for online DeepSIC and +0.184 for LMMSE; in the ten blocks after it, Citadel-Rx was back at its no-spoof SER (+0.0101) and online DeepSIC was not (+0.0674). Streak recovery was attempted 13 times in these runs and kept once, changing 1 of 6 runs; it had no measurable effect, and a denial attack that would test it, one that trips the gates on every block, was not found.

E. Retention Under Optimised Slow Drift

Table XII and Fig. 3 give the paired displacement for the A1 runs with level snapshots. The largest displacement of the slowest level in the window is 4.11 at SIR 0 dB and 3.78 at 10 dB (means over seeds), below the structural bound (15.4 and 13.5) and the design bound (18.0). This displacement is not a footprint the attack left in the memory. Over the window the slowest level of the clean run moved by 8.4 and that of the attacked run by 4.8, in nearly the same direction (cosine 0.95 between the two movements): rejected blocks enter the consolidation with weight $\gamma_b \approx 0$, so the attacked memory follows the channel more slowly, by 58% of the clean movement at SIR 0 dB and 78% at 10 dB, and D_K measures that lag.

The A3* runs (400 blocks each) were not complete when this version was compiled; their rows of Table XII appear when `revision_numbers.py` is re-run on their records.

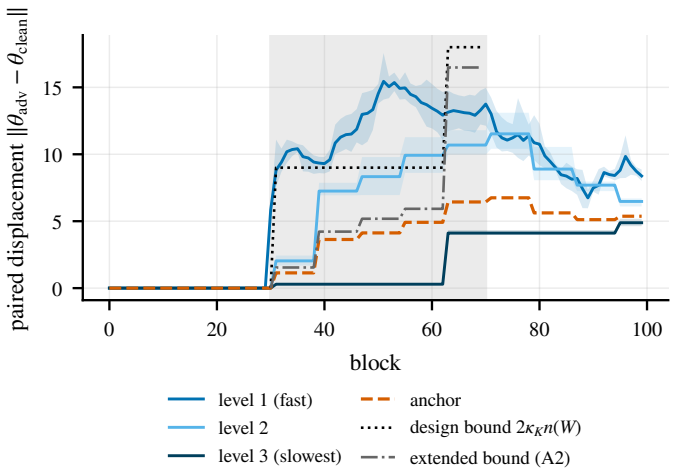


Fig. 3. Paired displacement of every level and of the anchor under A1 at SIR 0 dB, window 30–69 shaded (mean over seeds 10–12, band from the smallest to the largest seed), with the design bound (7) and the structural bound (8) (largest over the seeds).

TABLE XIII
FURTHER BASELINES UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11.

Receiver	Seeds	Clean SER	Excess SER [95% CI]	Window SER
LMMSE	2	0.0561	+0.298 [0.243, 0.345]	0.345
Robust LMMSE (Huber)	2	0.0561	+0.258 [0.207, 0.305]	0.305
Rollback, three gates	2	0.0331	+0.224 [0.152, 0.287]	0.251
L2-SP	2	0.0384	+0.754 [0.659, 0.838]	0.784
EMA teacher	2	0.0285	+0.414 [0.247, 0.584]	0.436
EWC	2	0.0353	+0.763 [0.678, 0.832]	0.791
CRC-verified DD	2	0.0262	+0.739 [0.596, 0.866]	0.760
CoTTA-style	2	0.0262	+0.241 [0.157, 0.332]	0.261
EATA-style	2	0.0353	+0.763 [0.678, 0.832]	0.791
Drift detection	2	0.0746	+0.700 [0.673, 0.723]	0.767
Single memory (Titans)	2	0.0556	+0.742 [0.712, 0.767]	0.786
Median-of-means gradient	2	0.0676	+0.682 [0.652, 0.713]	0.741
L2-SP, weight 0.001	2	0.0370	+0.780 [0.680, 0.864]	0.809
L2-SP, weight 0.1	2	0.0390	+0.600 [0.529, 0.673]	0.629
L2-SP, weight 1	2	0.0398	+0.124 [0.0888, 0.164]	0.154

F. Other Baselines

None of the further baselines resists A1 (Table XIII); at their default settings their excess SERs lie between +0.224 and +0.763. Robust statistics do not help against a perturbation spread over all pilots of a block: Huber-robust LMMSE +0.258, median-of-means gradient +0.682. Anchoring to the pre-trained weights trades protection for adaptation: L2-SP limits the damage in proportion to its weight, down to +0.124 at weight 1, the smallest excess of any baseline, with a clean SER of 0.0398. The EATA-style filter never fired, because the attacked receiver is confidently wrong: its mean data entropy stayed at or below 0.37 nats against a threshold of 0.55 nats, so the EATA-style receiver coincides with EWC. CRC-verified decision-directed training still adapts on the poisoned pilots (+0.739); the CoTTA-style teacher with stochastic restore keeps +0.241, and the single-memory receiver with the fast learner of Citadel-Rx and no gates +0.742, so the fast learner is not what protects Citadel-Rx.

TABLE XIV
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11.

Configuration	Seeds	Clean SER	Excess SER [95% CI]	Rej.	False rej.
Complete receiver	2	0.0159	$+0.0143$	0.988	0.000
$K = 1$ (gates, rollback, no slow level)	2	0.0346	$+0.404$ [0.312, 0.479]	0.950	0.100
$K = 2$, $C = (1, 8)$	2	0.0173	$+0.0251$ [0.0171, 0.0364]	0.975	0.000
No gates	2	0.0159	$+0.170$ [0.0996, 0.252]	0.000	0.000
G1 only	2	0.0158	$+0.0171$ [0.0097, 0.0261]	0.963	0.000
G2 only	2	0.0158	$+0.0146$ [0.0084, 0.0220]	0.988	0.000
G3 only	2	0.0158	$+0.162$ [0.0881, 0.245]	0.138	0.000
No anchor ($\lambda = 0$)	2	0.0229	$+0.0077$ [0.0053, 0.0103]	1.000	0.142
No fallback	2	0.0159	$+0.0705$ [0.0349, 0.113]	0.988	0.000
No CRC selection	2	0.0321	$+0.0013$ [-0.0021, 0.0066]	0.988	0.000
No consolidation clip	2	0.0159	$+0.0143$ [0.0083, 0.0218]	0.988	0.000

G. Ablations

Table XIV removes one part at a time; on seeds 10–11 the complete receiver has an excess of $+0.0143$ and a clean SER of 0.0159. Without the slow levels ($K = 1$, gates, rollback and fallback to the last accepted weights kept) the excess is $+0.404$: the receiver fails like CRC-Rollback, for the reason of Section VI-B, and it also rejects 0.100 of the clean blocks. Two levels, $C = (1, 8)$, give $+0.0251$. Without the fallback, a rejected block is still reverted but its data are detected with the rejected update: $+0.0705$. Without gates the excess is $+0.170$. Either decision-level gate suffices on its own here (G1 $+0.0171$, G2 $+0.0146$); the channel-prior gate G3 does not ($+0.162$, rejecting 0.138 of the poisoned blocks). Removing the proximal pull to the anchor lowers the excess to $+0.0077$ only because the clean twin is worse (0.0229) and rejects 0.142 of its blocks. Without the CRC selection the clean SER doubles (0.0321) and the excess nearly vanishes ($+0.0013$): the selection is most of what the receiver gives up while it rejects. The clip bound in 5/150 consolidations of the clean twins and 12/150 of the attacked runs at the headline cell, and removing it leaves the excess unchanged ($+0.0143$); it provides the guarantee of Theorem 1(i), not the measured robustness.

H. Block Length, Arrays, Modulation and Cost

Shorter blocks do not break the defence (Table XV): with $N_d \in \{1000, 5000\}$ and $N_p \in \{32, 100\}$ the excess of Citadel-Rx stays between $+0.0138$ and $+0.0170$, with at least 0.963 of the poisoned blocks rejected, against $+0.629$ to $+0.758$ for online DeepSIC. Fewer CRC sub-blocks make the pass fractions noisier, and the clean false rejection rises to 0.042 at $N_d = 1000$, $N_p = 32$. With 4×16 antennas the excess is $+0.0142$, against $+0.772$ for online DeepSIC and $+0.164$ for LMMSE.

Citadel-Rx fails at 16-QAM and 20 dB. There the pre-trained network passes a fraction 0.12 of the CRCs on clean blocks and the LMMSE reference 0.93, so the relative gates reject 97% of the clean blocks, the network hardly ever adapts, and the clean SER (0.120) is worse than that of online DeepSIC (0.0521) and of LMMSE (0.0094). The CRC selection keeps the gate's choice in the sub-blocks where nothing passes, so it does not recover the better LMMSE decisions, and under attack the receiver is no better protected ($+0.512$). The relative gates presuppose a network that is at least as good as the model-based reference on clean blocks; when it is not, a persistent clean false rejection is the symptom, and falling back to the reference is the obvious remedy, not evaluated here.

The two cheaper variants keep the protection: 30 fast steps per block give an excess of $+0.0171$ at 1.93 s per block, but reject 0.132 of the clean blocks, and the last-layer variant gives $+0.0145$ at 2.65 s. The full receiver takes 3.03 s per block, against 0.56 s for tuned online DeepSIC, 16.33 s for Meta-DeepSIC and 0.03 s for LMMSE: it takes 300 full-batch steps per stage and block where tuned online DeepSIC takes 30, and detects the data with both the fast level and the anchor.

TABLE XV
BLOCK LENGTH, A 4×16 ARRAY, 16-QAM AND TWO CHEAPER VARIANTS, WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11; TIME PER BLOCK ON ONE CORE OF THE LAPTOP.

Setting	Receiver	Seeds	Clean	Excess SER [95% CI]	Rej.	s/ block
$N_d = 1000$, $N_p = 32$	Citadel-Rx	2	0.0211	$+0.0160$ [0.0088, 0.0246]	0.963	1.3
$N_d = 1000$, $N_p = 32$	Online DeepSIC	2	0.0382	$+0.629$ [0.454, 0.770]	–	0.1
$N_d = 1000$, $N_p = 100$	Citadel-Rx	2	0.0164	$+0.0140$ [0.0079, 0.0217]	0.988	1.6
$N_d = 1000$, $N_p = 100$	Online DeepSIC	2	0.0342	$+0.756$ [0.631, 0.863]	–	0.2
$N_d = 5000$, $N_p = 32$	Citadel-Rx	2	0.0198	$+0.0170$ [0.0099, 0.0259]	0.963	1.5
$N_d = 5000$, $N_p = 32$	Online DeepSIC	2	0.0372	$+0.648$ [0.492, 0.774]	–	0.2
$N_d = 5000$, $N_p = 100$	Citadel-Rx	2	0.0160	$+0.0138$ [0.0079, 0.0211]	0.988	1.9
$N_d = 5000$, $N_p = 100$	Online DeepSIC	2	0.0334	$+0.758$ [0.650, 0.855]	–	0.2
$U \times N = 4 \times 16$	Citadel-Rx	2	0.0173	$+0.0142$ [0.0075, 0.0223]	1.000	3.6
$U \times N = 4 \times 16$	Online DeepSIC	2	0.0335	$+0.772$ [0.633, 0.886]	–	0.4
$U \times N = 4 \times 16$	LMMSE	2	0.0573	$+0.164$ [0.142, 0.183]	–	0.0
16-QAM, 20 dB	Citadel-Rx	2	0.120	$+0.512$ [0.401, 0.633]	0.787	5.5
16-QAM, 20 dB	Online DeepSIC	2	0.0521	$+0.941$ [0.915, 0.961]	–	0.5
16-QAM, 20 dB	LMMSE	2	0.0094	$+0.950$ [0.945, 0.955]	–	0.1
30 fast steps	Citadel-Rx	2	0.0210	$+0.0171$ [0.0105, 0.0259]	0.975	1.9
last layer adaptable	Citadel-Rx	2	0.0190	$+0.0145$ [0.0083, 0.0222]	0.988	2.6

TABLE XVI
ONE-AT-A-TIME SENSITIVITY, WHITE-BOX A1 AT SIR 0 DB, TEST SEED 10: CLEAN SER, EXCESS SER AND CLEAN FALSE REJECTION WITH EACH SETTING HALVED AND DOUBLED.

	$\times 0.5$			$\times 2$		
Setting	Clean	Excess	False rej.	Clean	Excess	False rej.
G1 slack $\tau_1 - 1$	0.0158	$+0.0140$	0.000	0.0158	$+0.0144$	0.000
G3 slack $\tau_3 - 1$	0.0157	$+0.0134$	0.000	0.0159	$+0.0130$	0.000
tightening $1-$floor	0.0159	$+0.0133$	0.000	0.0158	$+0.0137$	0.021
soft-step width	0.0158	$+0.0129$	0.000	0.0158	$+0.0146$	0.000
proximal weight λ	0.0163	$+0.0142$	0.000	0.0163	$+0.0139$	0.000
delta decay δ	0.0156	$+0.0132$	0.000	0.0162	$+0.0131$	0.000
CRC relative slack ρ	0.0159	$+0.0130$	0.053	0.0158	$+0.0133$	0.000
level-3 rate $\eta^{(3)}$	0.0157	$+0.0136$	0.000	0.0159	$+0.0130$	0.000

I. Sensitivity

Halving or doubling any one of eight settings (Table XVI, seed 10) keeps the clean SER between 0.0156 and 0.0163, the excess between $+0.0129$ and $+0.0146$, and the clean false rejection at or below 0.053; no setting of the gates or of the memory is critical within a factor of two. The test uses one seed and two multipliers per setting, so it shows the absence of a cliff near the chosen values, not an optimum.

VII. Discussion

What the results support. A test-time receiver can be protected against pilot poisoning by an update rule alone, without detecting the attacker, if what it keeps is checked against something the attacker cannot reach. In Citadel-Rx that is the slow memory, through the anchor detector, and the CRC of the data; CRC-Rollback has the same test but judges against LMMSE on the poisoned pilots, and fails for that reason. What remains under attack, an excess of $+0.0188$ at the headline cell, is the price of rejecting: forty blocks detected with a memory that did not adapt to them. The gain without attack is mostly the CRC selection, which any receiver with a CRC can use.

Where it fails. The defence is only as good as its references. An attacker that also perturbs the data degrades them and gets more poisoned blocks accepted (A4), and the abstain state designed for this case locks on clean runs as implemented. Where the network is worse than LMMSE, as at 16-QAM here, the relative gates reject nearly every clean block and the receiver is worse than both alternatives. After an abrupt legitimate change the gates slow the re-adaptation. The gates do not address jamming, which at the same SIR harms every receiver more than poisoning harms Citadel-Rx.

Scope. The evaluation is link-level and uncoded apart from the CRC: there is no NR slot, no LDPC code and no attacker placed in a ray-traced scene; the SIR is translated to distance analytically. The test channel is the SED model with cosine fading (COST2100 traces enter the calibration only), the backbone is DeepSIC only, the longest runs are 400 blocks, and per-user gating is implemented but not evaluated. The headline cell has ten test seeds, the other experiments two or three, so their intervals are wide and seed-level tests are possible only at the headline cell. The theory bounds the displacement of the slowest level, not the error rate; the design bound holds for any attacker but is coarse, and the structural bound needs the paired clean run, so it is a diagnostic of an experiment, not a run-time certificate. All results of the first version used the consolidation rule corrected in Section III and are superseded by this evaluation. Times were measured on laptop and cloud CPUs with several runs in parallel; no GPU measurement is reported.

VIII. Conclusion

A test-time neural receiver resists pilot poisoning when it consolidates into persistent memory only what a label-free physical check verifies, and when the check judges every update against that memory. Citadel-Rx realises this with gated, clipped consolidation into two slow levels, a fallback to the slow memory and per-sub-block CRC selection, and a clip that bounds how far any attacker can move the slowest level. On held-out seeds, white-box poisoning at SIR 0 dB raised its SER by $+0.0188$ against $+0.768$ for tuned online DeepSIC and $+0.271$ for CRC-Rollback, which has the same test but no protected reference; attacks built against the gates did not raise this excess. The open problems are the ones the evaluation exposed: references an attacker cannot degrade through the data, a reference model for abstaining that follows legitimate change, a fallback for networks weaker than the model-based detector, and a validation in a coded NR link.

Acknowledgment

The COST2100 traces are those released with [6].

References

[1] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.

[2] M. Honkala, D. Korpi, and J. M. J. Huttunen, "DeepRx: Fully convolutional deep learning receiver," *IEEE Trans. Wireless Commun.*, vol. 20, no. 6, pp. 3925–3940, Jun. 2021.

[3] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, May 2019.

[4] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.

[5] N. Shlezinger, R. Fu, and Y. C. Eldar, "DeepSIC: Deep soft interference cancellation for multiuser MIMO detection," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1349–1362, Feb. 2021.

[6] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.

[7] K. Wu, W. Jiang, D. Niyato, Y. Li, and C. Luo, "Transfer-based adversarial poisoning attacks for online (MIMO-)deep receviers," arXiv:2409.02430, Sep. 2024.

[8] X. Zhou, B. Maham, and A. Hjørungnes, "Pilot contamination for active eavesdropping," *IEEE Trans. Wireless Commun.*, vol. 11, no. 3, pp. 903–907, Mar. 2012.

[9] D. Kapetanović, G. Zheng, and F. Rusek, "Physical layer security for massive MIMO: An overview on passive eavesdropping and active attacks," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 21–27, Jun. 2015.

[10] D. Kapetanović, G. Zheng, K.-K. Wong, and B. Ottersten, "Detection of pilot contamination attack using random training and massive MIMO," in *Proc. IEEE 24th Annu. Int. Symp. Pers., Indoor, Mobile Radio Commun. (PIMRC)*, 2013, pp. 13–18.

[11] J. K. Tugnait, "Self-contamination for detection of pilot contamination attack in multiple antenna systems," *IEEE Wireless Commun. Lett.*, vol. 4, no. 5, pp. 525–528, Oct. 2015.

[12] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.

[13] S. Schibisch, S. Cammerer, S. Dörner, J. Hoydis, and S. ten Brink, "Online label recovery for deep learning-based communication through error correcting codes," in *Proc. 15th Int. Symp. Wireless Commun. Syst. (ISWCS)*, 2018, pp. 1–5.

[14] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 162, 2022, pp. 16888–16905.

[15] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 7201–7211.

[16] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[17] L. Yuan, B. Xie, and S. Li, "Robust test-time adaptation in dynamic scenarios," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15922–15932.

[18] T.-H. Hoang, D. M. Vo, and M. N. Do, "Persistent test-time adaptation in recurring testing scenarios," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024.

[19] X. Li, Y. Grandvalet, and F. Davoine, "Explicit inductive bias for transfer learning with convolutional networks," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2825–2834.

[20] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[21] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 1195–1204.

[22] T. Wu, F. Jia, X. Qi, J. T. Wang, V. Sehwag, S. Mahloujifar, and P. Mittal, "Uncovering adversarial risks of test-time adaptation," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 202, 2023, pp. 37456–37495.

[23] H. Park, J. Hwang, S. Mun, S. Park, and J. Ok, "MedBN: Robust test-time adaptation against malicious test samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 5997–6007.

[24] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 119–129.

[25] J. Steinhardt, P. W. Koh, and P. Liang, "Certified defenses for data poisoning attacks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 3517–3529.

[26] A. Behrouz, M. Razaviyayn, P. Zhong, and V. Mirrokni, "Nested learning: The illusion of deep learning architectures," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2512.24695.

[27] A. Behrouz, P. Zhong, and V. Mirrokni, "Titans: Learning to memorize at test time," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2501.00663.

[28] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly, "Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory," *Psychol. Rev.*, vol. 102, no. 3, pp. 419–457, Jul. 1995.

[29] D. Kumaran, D. Hassabis, and J. L. McClelland, "What learning systems do intelligent agents need? Complementary learning systems theory updated," *Trends Cogn. Sci.*, vol. 20, no. 7, pp. 512–534, Jul. 2016.

[30] M. Sadeghi and E. G. Larsson, "Adversarial attacks on deep-learning based radio signal classification," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 213–216, Feb. 2019.

[31] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 274–283.

[32] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, "Synthesizing robust adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 284–293.

[33] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," arXiv:1902.06705, 2019.

[34] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1633–1645.

[35] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2137–2146.

[36] D. Wierstra, T. Schaul, T. Glasmachers, Y. Sun, J. Peters, and J. Schmidhuber, "Natural evolution strategies," *J. Mach. Learn. Res.*, vol. 15, no. 27, pp. 949–980, 2014.

[37] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 70, 2017, pp. 1126–1135.

[38] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[39] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901, V19.5.0, 2026, Table 7.4.1-1.

[40] 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and channel coding," TS 36.212, V19.1.0, 2025, Sec. 5.1.1.

[41] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin, F. Tufvesson, and P. De Doncker, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.

[42] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, Mar. 1964.

[43] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *J. Amer. Statist. Assoc.*, vol. 58, no. 301, pp. 13–30, Mar. 1963.

[44] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Ann. Statist.*, vol. 17, no. 3, pp. 1217–1241, Sep. 1989.

[45] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. Amer. Statist. Assoc.*, vol. 22, no. 158, pp. 209–212, Jun. 1927.

[46] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, Dec. 1945.

[47] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.